```
void MmwDemo_transmitProcessedOutput(UART_Handle uartHandle,
                                     DPC_ObjectDetection_ExecuteResult
*result,
                                     MmwDemo_output_message_stats
*timingInfo)
{
    MmwDemo_output_message_header header;
    uint32_t tlvIdx = 0;
    uint32_t packetLen;
    uint8_t padding[MMWDEMO_OUTPUT_MSG_SEGMENT_LEN];
    uint32_t numPaddingBytes;

    DPC_ObjectDetection_PreStartCfg objDetPreStartCfg;
    DPC_ObjectDetection_StaticCfg *staticCfg =
&objDetPreStartCfg.staticCfg;

    MmwDemo_output_message_tl   tl[MMWDEMO_OUTPUT_MSG_MAX];

    cmplx16ImRe_t *radarcube = (cmplx16ImRe_t*)result->radarCube.data;

    /* Clear message header */
    memset((void *)&header, 0, sizeof(MmwDemo_output_message_header));
    /* Header: */
    header.platform =  0xA6843;
    header.magicWord[0] = 0x0102;
    header.magicWord[1] = 0x0304;
    header.magicWord[2] = 0x0506;
    header.magicWord[3] = 0x0708;
    header.numDetectedObj = result->numObjOut;
    header.version =    MMWAVE_SDK_VERSION_BUILD |   //DEBUG_VERSION
                        (MMWAVE_SDK_VERSION_BUGFIX << 8) |
                        (MMWAVE_SDK_VERSION_MINOR << 16) |
                        (MMWAVE_SDK_VERSION_MAJOR << 24);

    packetLen = sizeof(MmwDemo_output_message_header);

    /* Add RadarCube TLV */
    tl[tlvIdx].type = MMWDEMO_OUTPUT_EXT_RADAR_CUBE;
    tl[tlvIdx].length = result->radarCube.dataSize;
    packetLen += sizeof(MmwDemo_output_message_tl) + tl[tlvIdx].length;
    tlvIdx++;
```

```
    /* Fill header */
    header.numTLVs = tlvIdx;
    /* Round up packet length to multiple of MMWDEMO_OUTPUT_MSG_SEGMENT_LEN
*/
    header.totalPacketLen = MMWDEMO_OUTPUT_MSG_SEGMENT_LEN *
            ((packetLen +
(MMWDEMO_OUTPUT_MSG_SEGMENT_LEN-1))/MMWDEMO_OUTPUT_MSG_SEGMENT_LEN);
    header.timeCpuCycles =  Pmu_getCount(0);
    header.frameNumber = result->stats->frameStartIntCounter;
    header.subFrameNumber = result->subFrameIdx;

    /* Send header */
    UART_writePolling (uartHandle,
                       (uint8_t*)&header,
                       sizeof(MmwDemo_output_message_header));

    /* Send RadarCube TLV */
        UART_writePolling(uartHandle,
                          (uint8_t*)&tl[0],
                          sizeof(MmwDemo_output_message_tl));

        /* Send RadarCube data in DPIF_RADARCUBE_FORMAT_1 */
        uint32_t numTXPatterns =staticCfg->numTxAntennas;
        uint32_t numDopplerChirps = staticCfg->numDopplerChirps;
        uint32_t numRX = staticCfg->ADCBufData.dataProperty.numRxAntennas;
        uint32_t numRangeBins = staticCfg->numRangeBins;




        uint32_t tx;
        uint32_t chirp;
        uint32_t rx;
        uint32_t index = 0;

        for ( tx = 0; tx < numTXPatterns; tx++)
        {
            for (chirp = 0; chirp < numDopplerChirps; chirp++)
            {
                for (rx = 0; rx < 1; rx++)
                {
                    //uint32_t index  = tx * numDopplerChirps * numRX *
numRangeBins + chirp * numRX * numRangeBins + rx * numRangeBins;
```

```
                    UART_writePolling(uartHandle,
                                      (uint8_t*)&radarcube[index],
                                      sizeof(cmplx16ImRe_t) * 1);
                    index += chirp*numRX*numRangeBins;


                }
            }
        }



    /* Send padding bytes */
    numPaddingBytes = MMWDEMO_OUTPUT_MSG_SEGMENT_LEN - (packetLen &
(MMWDEMO_OUTPUT_MSG_SEGMENT_LEN-1));
    if (numPaddingBytes < MMWDEMO_OUTPUT_MSG_SEGMENT_LEN)
    {
        UART_writePolling(uartHandle,
                          (uint8_t*)padding,
                          numPaddingBytes);
    }
}
```